

What an LLM Judge Cannot See in Long-Form Generation: Injected Text, Self-Copy and the Horizon

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Abstract

Evaluating thousands of tokens of model-generated text with an LLM judge that reads one window at a time is common practice. In a study of interventions on long, open-ended generation by base language models, judged by Claude Opus 5 on 96- to 160-token windows with 600 tokens of context, we found three things the judge could not see, each large enough to change the conclusions. (1) Text the experimenter injects into a stream (a new subject every few hundred tokens) ends up inside the judged windows and is scored as the model’s own: a reported growth of the effect with the period was mostly the injected sentence. (2) Given the same four injected sentences in rotation, a base model replays its own earlier segments from 600 to 1,200 tokens back, beyond the judge’s horizon; 65–80% of post-interruption windows at periods 150–300 were more than half verbatim reproductions, and the judge scored them as surprise and, worse, as connection. (3) Local gains do not compose: on whole documents no condition built an integrated text, and the condition with the best windows read as a sequence of restarts. We describe the protocol that corrects these (windows of generated text only, fresh-only estimates with a shingle-based self-copy flag, a document-level reading, the premise rather than the window as the unit of inference, a pipeline test–retest from two byte-identical arms, a second judge family and human readers) and what survives it: the intervention’s effect on surprise is real and about one point smaller; its effect on connection was mostly artefact. The lesson is general: a windowed judge measures what is in the window, and what a long stream does is partly outside it.

1 Setting

The underlying study [1] asks where the novelty a base language model produces with no task comes from. A cell is one premise (a single sentence) continued by a base model (mainly Qwen3-30B-A3B-Base, 8-bit, MLX) for 4,500 tokens with the end-of-text token masked; arms differ in whether a windowed repetition penalty (*habituation*) is on and in whether, how often and with what the stream is *interrupted*: a short sentence injected into the model’s cache as if it had written it, such as “*Meanwhile, in a city with no name,*”, every 150 to 900 tokens. Ten premises are shared by all arms; 24 arms were run. Windows of the stream are judged by Claude Opus 5 on three 0–10 dimensions (surprise given the earlier text, connection to it, coherence), five calls per window with the median as the score, the 600 preceding tokens shown as context. The judge was measured: intra-window spread 0.45–0.70, agreement with a second family (Kimi K2.6) $\rho = 0.65$ to 0.85, agreement of three human raters’ consensus with the judge $\rho = 0.58$ on surprise. The first version of the study, written under this protocol, reported that the interruption raised surprise from 1.7 to 3.1 and connection from 1.1 to 3.2 over habituation, that its yield grew with the length of the thread it broke (5.3–5.9 after 300–900 tokens against 3.1 after 150), and that a period of about 300 tokens was best. Each of these numbers was partly an artefact of what the judge could and could not see.

arm (Qwen3-30B-A3B-Base)	windows	copied	source beyond the judge's 600 tokens	fresh-only S / C / H
bare	60	67%	0%	1.77 / 1.38 / 5.55
habituation	60	27%	8%	1.76 / 1.32 / 5.01
habituation + interruption 150	60	72%	12%	2.95 / 2.03 / 6.78
habituation + interruption 300	60	65%	62%	3.02 / 2.03 / 6.52
habituation + interruption 600	60	38%	35%	2.84 / 2.04 / 6.33
habituation + interruption 900	50	12%	12%	3.12 / 2.25 / 6.75
sham break 300	60	32%	5%	2.16 / 1.41 / 5.03
reset + new subject 300	60	0%	0%	3.75 / 3.30 / 6.87

Table 1: Self-copy in the judged (post-interruption or grid) windows: share of windows that are at least half verbatim reproductions of the stream’s own earlier text, share whose source lies beyond the 600 tokens shown to the judge, and cell means of surprise / connection / coherence on the fresh windows only (ten premises).

2 Three things the judge could not see

1. The injected text. The first protocol cut 160-token windows right before each injection and on a uniform grid, so that a window after an interruption contained the injected sentence. The judge, asked how unexpected the window is given the earlier text, read the experimenter’s sentence as the model’s turn. Re-judging with windows of generated text only (96 tokens starting 32 tokens after the end of the injected sentence, none crossing an injection; uniform grid in uninterrupted arms) kept the direction of every effect and removed the period story entirely: the post-interruption window scores 3.0 at period 150, 2.9 at 300, 2.9 at 600 and 3.2 at 900, every paired difference covering zero. Across 14 conditions the means under the two protocols correlate at $\rho = +0.82$ (surprise) to $+0.94$ (connection); the sizes differ exactly where an injected sentence sat in the window.

2. Self-copy beyond the horizon. A base model given the same four subject-change sentences in rotation learns the rotation: after “*That night she dreamed of something else entirely.*” it tends to write what it wrote the last time that sentence appeared, 600 or 1,200 tokens earlier. We flag a judged window as *copied* when at least half of its 12-token shingles occur earlier in the same stream, and record whether the source lies within the 600 tokens the judge sees (Table 1). Bare generation is 67% copied (literal loops, all visible to the judge, who scores them 0.0); habituation alone 27%; the interruption arms with the fixed rotation 65–80% at periods 150 and 300, and at 300 the source of 62% of all windows lies outside the judge’s context; 38% at 600, 12% at 900; the arm whose context is reset at each interruption 0%. A copied window scores high on “connection” for the obvious reason (it *is* earlier text) and, when its source is out of view, on surprise. On fresh windows only, the interruption still adds $+1.2$ [$+0.4, +2.1$] of surprise at period 150 and $+1.3$ at 300 over habituation ($p \leq 0.01$, exact paired permutation over ten premises), but its connection effect falls from $+2.4$ to $+0.75$; the reset arm, which cannot replay, adds $+1.9$ and $+2.0$. We found this because a document-level judge (below) remarked that the stream “repeats the same four-part block verbatim”.

3. Windows and wholes. Every score above is local. We had the judge read the whole 4,500-token stream of each cell, injected sentences removed, and score integration (parts taken up and joined later), development (something builds rather than restarts), coherence and surprise once per document ($k = 3$). No condition exceeds 2.5 on any dimension. The interrupted stream over the preserved context, whose windows are the best, reads *below* the uninterrupted habituated one (surprise 0.1 vs 1.4, $p = 0.002$; development 0.3 vs 1.1): a sequence of restarts. The reset arm reads best (2.0 / 1.6 / 2.2 / 2.5), because each of its parts is a clean piece that returns to

the premise; this replicated on ten new premises.

3 The protocol that corrects them

1. **Windows of generated text only**, starting a margin after any injected text and never crossing an injection; the context shown to the judge may contain injected text, the graded region never does.
2. **A self-copy flag** per window (shingles seen earlier in the stream; source within or beyond the judge’s horizon) and **fresh-only estimates** reported alongside the all-windows ones; a design rule that follows: never inject the same sentence twice, or reset the context.
3. **A document-level reading** of the whole stream, with injected text removed, as the check that local gains compose.
4. **The stream, not the window, as the unit of inference**: cell means, bootstrap intervals over cells, exact sign-flip permutation tests on paired cells, effect sizes on cell means; window-level statistics only as descriptive.
5. **A measured judge**: the median of k calls; the intra-window spread; a pipeline test–retest (two arms that turned out byte-identical in all ten cells were judged independently: medians of five agree exactly in 91% of windows on surprise, cell means differ by -0.01 $[-0.09, +0.07]$); a second judge family on a random, not judge-stratified, sample ($\rho = 0.75 / 0.80 / 0.65$ on surprise / connection / coherence); and human raters on fresh windows.
6. **A pre-registered confirmatory battery** on new premises and a new random seed, analyzed under the protocol fixed in advance: the primary contrast confirmed ($p = 0.003$; on fresh windows $p = 0.04$), the expectation that a kept context carries connection refuted.

4 What survives

The intervention is real and smaller than first reported: a new subject injected every few hundred tokens into a habituated stream raises judged surprise on fresh generated text by about 1.2 to 1.4 points over habituation alone, on three base models from two families, a post-trained model, an unquantized model and a second genre; a bare paragraph break does nothing; the kept context is not what carries the effect, and the connection it seemed to carry was mostly replay. What did not survive were the claims that depended on what the judge could not see. We think the general point holds for any evaluation of long generation with a windowed LLM judge: the judge measures what is in the window, and a long stream does much of what it does outside it, including copying the experimenter and copying itself. Code, data, prompts and the analysis are in the repository of Ono Filho [1].

References

- [1] Roberto I. Ono Filho. Interrupting the loop: Periodic subject changes raise judged surprise and connection in base language models. Preprint, 2026. ORCID 0009-0006-8650-629X.